

Do Superseded States Change Language-Model Decisions? A Controlled Test Against Incidental Mentions

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Abstract

A model that reads an updated record can report the current value correctly and still let the old value affect what it decides. We test whether that happens, and whether it happens *because* the old value was an assignment that has since been superseded. In synthetic access-control and routing records, the current state, the decision policy and the correct action are held fixed while a single obsolete state is edited so that it implies either the correct or the wrong action. Matched controls place the same word as an assignment to an unrelated attribute that was later updated, as a word the same entity merely mentioned, or as an unattached word. In a frozen confirmation on held-out wording (Qwen3-8B, 768 histories), the obsolete state raised the wrong action’s log-odds by 3.904 and 3.133 nats in the two tasks and flipped the generated decision in about one history in six, almost always toward the wrong action (65 against 1 and 75 against 4 histories), including in routing conditions where the model retrieved both the current (0.982–0.988) and the initial (0.990) state and decided correctly without history (0.953–0.961). The effect is not specific to supersession: an incidental mention of the same word by the same entity moved decisions as much in access and more in routing (net switches 0.294 against 0.185), and whether a superseded assignment exceeded the mention control in score depended on the prompt template. A superseded assignment exceeded an updated unrelated attribute in routing but not detectably in access. Finally, the edit’s effect on easy retrieval queries, a score-level sensitivity measure, added no information about which decisions would fail beyond simple pre-decision predictors (log-loss gain +0.0004 nats, 95% interval [−0.0011, +0.0019]). A second model, Gemma 3 4B, did not meet the task-competence gate, so the results are for one model.

1 Introduction

Agents and assistants increasingly act on records that change: a user’s clearance is revoked, a ticket is rerouted, a preference is replaced. The record usually keeps the history, and the model is told that only the latest entry counts. Retrieval benchmarks ask whether the model can then report the current value, and recent work shows that old values do intrude on such reports, especially after many overwrites (Wang and Sun, 2025; Guo et al., 2026; Chat-taraj and Raj, 2026). A deployed system, however, rarely asks the model to restate a value. It asks the model to *use* it. Two questions follow that retrieval accuracy does not answer. When a model can state the current value correctly, does the obsolete value still change the decision it takes? And if it does, is the obsolete value acting as a superseded assignment of the queried attribute, or simply as a word that occurs in the context near the entity?

The second question matters because the remedies differ. If the interference is tied to supersession, it is a failure of state tracking, and update-aware memory, historical marking or attention routing toward the current entry (Guo et al., 2026) are the natural fixes. If any nearby mention of a decision-relevant word does the same, the problem is contextual entrainment (Niu et al., 2025) reaching decisions. Removing or marking stale entries would then leave most of it in place, and evaluations that only vary update histories would misattribute it.

Design. We hold everything that should determine the decision fixed: the current state, the policy that maps states to actions, the query, and the correct action. We edit one obsolete state. In one member of each pair, the obsolete state implies the wrong action; in the other, it implies the correct action. The difference in the model’s preference for the wrong action between the two members is the decision effect of the obsolete content, and

the paired generated answers give its behavioral counterpart (Figure 1). The same edit is repeated in matched constructions that keep the word and its position but change its role: an assignment to a different attribute that was later updated, a word the entity only mentioned, and a word attached to no entity. Before testing decisions, we check on the same records that the model reports the current and the initial state correctly when asked.

Findings. The experiment ran under a protocol frozen before the confirmation data were generated, on a held-out prompt template, with one model (Qwen3-8B) that passed a task-competence gate. A second model, Gemma 3 4B, did not pass it. Three conclusions hold for the confirmed model:

1. **Obsolete states change current decisions in a model that tracks them correctly.** Making the obsolete state imply the wrong action raised the wrong action’s log-odds by 3–4 nats and flipped roughly one generated decision in six toward it. Reverse flips were rare. In routing, this happened in every condition where the model reported the current and initial states with 0.98–0.99 accuracy and decided correctly without history (§5.2).
2. **The interference does not require supersession.** The same word mentioned in passing by the same entity moved decisions at least as much as a superseded assignment, and more in routing. Whether a superseded assignment exceeded the mention control at the score level changed with the prompt template (§5.3).
3. **A score-level sensitivity measure does not identify vulnerable items.** The edit’s effect on easy retrieval queries was large but did not predict which decisions would fail beyond simple pre-decision features (§5.4).

What is and is not new. That stale context interferes with answers is established (Wang and Sun, 2025; Guo et al., 2026; Qian et al., 2026; Chao et al., 2026), as is the fact that mentioned tokens gain probability (Niu et al., 2025). Our contribution is a controlled test that connects the two at the level of decisions. We measure interference on policy decisions in an open model, on items whose current and historical states the model retrieves correctly. We use minimal-pair edits that hold the correct action fixed, report both scores and generated answers, and include the matched controls

needed to ask whether supersession is the operative factor. We also report the negative result for a sensitivity-based predictor and, in Appendix A, two measurement problems we found along the way: chat-format events that the model never generates, and log-probabilities that change when tokens are appended under 8-bit and bfloat16 inference.

2 Related work

Interference from updated values. Proactive interference grows with the number of overwrites of a key, and errors are mostly earlier values of the same key (Wang and Sun, 2025). Models also retrieve the first value of a key more reliably than the latest (Chattaraj and Raj, 2026; Qiao et al., 2026). Guo et al. (2026) name the failure *stale binding*. They find that almost all errors on current value queries return an old value of the queried variable, that the current value stays linearly recoverable on failures, that moving a historical mention near the query increases errors and historical wording reduces them, and that test-time attention routing repairs most errors in 3–9B models. Their decision scenarios, run with four closed reasoning models, showed no difference between full histories and resolved snapshots. Qian et al. (2026) and Chao et al. (2026) evaluate forgetting and stale memories in dialogue and agent settings. STALE in particular separates recognizing an update from applying it in a later policy. Both are accuracy benchmarks with naturalistic text. Our setting differs in three ways: a single update instead of long overwrite chains, decisions whose correct action is fixed by an explicit policy, and matched minimal pairs with lexical controls. The failures we study sit below the interference loads at which retrieval breaks.

Entrainment and distraction. Tokens that appear in the context gain probability whether or not they are relevant, random tokens included (Niu et al., 2025). Irrelevant context also degrades reasoning (Shi et al., 2023). Our entity-mention and unattached-word constructions measure this effect in the update setting. The question we ask is whether a superseded assignment adds decision pressure beyond it. Yang et al. (2026b) argue that counterfactual prompt edits are compound treatments and must be compared against matched alternative edits rather than against no edit. Our design follows that logic: each construction applies the same lexical edit, and the contrasts compare constructions.

Read the dated record. Only the final assignment governs the decision.
Initially, EvbUnit0096A had routing state pending. <- edited: pending | suspended
Initially, EvbUnit0096B had routing state suspended.
Later, EvbUnit0096A had archive tag restricted.
Finally, EvbUnit0096A had routing state restricted.
Later, EvbUnit0096B had archive tag restricted.
Finally, EvbUnit0096B had routing state pending.
Policy: state to bin: restricted -> B0, suspended -> B1, pending -> B2,
active -> B3; bin to action: B0 -> EAST, B1 -> EAST, B2 -> WEST, B3 -> WEST
Apply the supplied rule to the final assignment of EvbUnit0096A; give its action.
Respond with exactly one label (EAST, WEST), with no other text.

Construction	Edited line (same word, same position)
Superseded assignment	Initially, EvbUnit0096A had routing state pending.
Other attribute, updated	Initially, EvbUnit0096A had archive tag pending.
Mentioned by entity	Initially, EvbUnit0096A mentioned the word pending.
Unattached word	Initially, the unassigned word was pending.

Figure 1: One confirmation item (routing task, sequential order, held-out template; the annotation arrow is not part of the prompt). The current state restricted maps to EAST. The obsolete state is edited between pending (implies WEST, wrong) and suspended (implies EAST, correct); everything else is identical. The table shows how the edited line reads in each construction. The decision effect D_c compares the model’s preference for the wrong action across the two edits.

Binding and state tracking. Mechanistic work identifies how models bind attributes to entities (Feng and Steinhardt, 2024; Gur-Arieh et al., 2026; Prakash et al., 2026). It also shows that models aggregate state changes at query time rather than updating incrementally (Tang et al., 2026; Kim and Schuster, 2023). These accounts offer plausible reasons why lexical identity and position could matter more than assignment history, but our experiments are behavioral and do not test them.

Knowledge conflicts. Our records contain an intra-context conflict with an explicit temporal resolution (Xu et al., 2024). Related benchmarks ask for the value valid at a given time (Yang et al., 2026a). That is our historical retrieval check, not our decision task.

3 Task, constructions and estimands

Records and decisions. Each history describes two entities. Each entity has an initial entry, an update of a separate *archive tag*, and a final assignment of the decision-relevant attribute (Figure 1). A policy maps four states to two actions, two states per action. In the *access* task, clearance maps directly to APPROVE or DENY. In the *routing* task, the routing state maps to a bin and the bin to EAST or WEST. The model is asked for the action implied by one entity’s final assignment and must answer with exactly one label. Two vocabularies are used: natural states (active, pending, restricted, suspended) and opaque states (S1–

S4). Natural labels carry prior meanings that the policy may contradict. Three orders define the cells: *sequential* (both initial entries, then both final blocks), *interleaved* (initial and final blocks alternate across entities), and *competing* (interleaved, plus two archived discussions that quote decision-relevant state words for the two entities). The factorial crosses task, cell, vocabulary, correct action, historical order, current order and entity orientation: 192 histories per replicate. State-to-label assignments rotate so that every state appears in every role. In every history the other entity’s current state implies the wrong action. Confusing the two entities is therefore always an error, but it is identical in both members of a pair and cancels in the paired contrasts below.

Constructions. For the queried entity, the obsolete line takes one of four forms, all with the same edited word in the same slot: *superseded* (the queried attribute was initially assigned the word and later reassigned); *updated other attribute* (the archive tag was initially the word and was later updated); *entity mention* (the entity mentioned the word, with no assignment); and *unattached* (the unassigned word was ...). Every history also contains a *current-only* item with no initial lines, which measures whether the model can apply the policy at all, and a *live* item that edits the final assignment itself, which shows the size of a legitimate change. The constructions share the policy, query, archive-tag update, final assignments and

surrounding text. They differ in syntax around the edited word, so the contrasts compare constructions as wholes, not isolated semantic features.

Decision effect. For a prompt x , let $L(x) = \log P(a_{\text{wrong}} | x) - \log P(a_{\text{correct}} | x)$, where each action event is the label exactly as the model writes it at the start of its turn, followed by the end-of-turn token. For construction c and history h , the two members differ only in the obsolete word: in member $x_{h,c}^-$ it implies the wrong action, and in $x_{h,c}^+$ the correct one. The decision effect is

$$D_c = \frac{1}{|H|} \sum_{h \in H} [L(x_{h,c}^-) - L(x_{h,c}^+)]. \quad (1)$$

$D_c > 0$ means the obsolete content pushes preference toward the action it implies. It is a preference shift in nats, not an error rate. The two hypothesis contrasts are $\Delta_{\text{semantic}} = D_{\text{superseded}} - D_{\text{updated other}}$, which asks whether the assignment’s attribute matters, and $\Delta_{\text{mention}} = D_{\text{superseded}} - D_{\text{entity mention}}$, which asks whether assignment adds anything to an incidental mention.

Generated switches. With greedy decoding, a history scores +1 for construction c if the model answers wrongly under x^- and not under x^+ , -1 for the reverse, and 0 otherwise. The mean over histories is the net switch rate.

Retrieval checks. On each record we also ask *easy* queries for the current state of the queried entity and of the other entity. For superseded items we ask for the *initial* state, under an introduction that announces both initial and final assignments. The answers are generated and parsed strictly.

Easy-query sensitivity. Let w^- and w^+ be the two values of the edited word, and let $m_x(w)$ be the log probability that easy query q , asked on record x , assigns to the answer w (summed over its spaced, unspaced and capitalized surface forms, scored as answer prefixes). Define $E(q) = [m_{x^+}(w^+) - m_{x^+}(w^-)] - [m_{x^-}(w^+) - m_{x^-}(w^-)]$. Then $R_{\text{easy}} = E(\text{queried entity}) - E(\text{other entity})$ is a per-history, query-specific sensitivity of retrieval scores to the edit. Hypothesis H3 asks whether R_{easy} helps predict downstream failures beyond features that are available before the decision query is asked.

4 Protocol

Model and runtime. Qwen3-8B (Yang et al., 2025) at a pinned revision (Appendix C), with thinking disabled through the chat template. Inference ran in float32 without quantization on one GPU. We chose float32 after finding that 8-bit and bfloat16 inference made the log probability at a fixed position change by up to 0.843 and 0.875 nats when tokens were appended after it. Float32 gave exactly zero (Appendix A.1). Action events were scored with a key-value-cache scorer. Before each run, the scorer was checked against one full forward pass per event on identical prompts (maximum difference 5.08×10^{-5} nats against a tolerance of 0.02). Answers were generated greedily with at most 16 new tokens and parsed strictly: an answer counts only if, after stripping outer whitespace, it is exactly one allowed label.

Splits and gate. Three disjoint splits were generated from separate seeds and separate entity-name pools. A development split (192 histories) was used only to check competence and answer formats. A frozen gate split (384 histories) decided eligibility. The confirmation split (768 histories, 8,448 records, 11 per history) used a held-out paraphrase of the introduction and query. A task-cell was *competent* if strict direct retrieval of the current state was at least 0.95, current-only decision accuracy at least 0.90, and invalid answers at most 0.05. Confirmation data were generated only after the gate, recomputed from its saved scores, showed at least one competent cell. All cells are reported. Task-level estimates average the fixed grid, and estimates from non-competent cells are labeled as such.

Hypotheses and statistics. The protocol fixed two primary hypotheses per task: H1, $D_{\text{superseded}} > 0$; and H2, $\Delta_{\text{semantic}} \neq 0$, two-sided. Both use 97.5% percentile intervals from a history bootstrap (2,000 draws, fixed seed), Bonferroni-adjusted over the two tasks. Secondary outcomes use unadjusted 95% intervals: Δ_{mention} , the other construction effects, net switch rates, and H3. Histories are the resampling unit. Both members, all constructions and all queries of a history stay together. The two tasks are never pooled, and an interval that includes zero is reported as inconclusive, not as evidence of no effect.

Cell	Direct	No history	History	Competent
access seq.	0.916	0.867	0.831	no
access int.	0.945	0.875	0.858	no
access comp.	0.908	0.859	0.867	no
routing seq.	0.988	0.953	0.858	yes
routing int.	0.986	0.961	0.868	yes
routing comp.	0.982	0.953	0.846	yes

Table 1: Strict accuracy on the confirmation data. *Direct*: current state of the queried entity, asked directly. *No history*: decision on the current-only item. *History*: decision over all historical-construction items. *Competent*: direct ≥ 0.95 , no history ≥ 0.90 , invalid ≤ 0.05 . No answer was invalid.

Prediction (H3). Predictors that are available before the decision query is asked: the easy query’s margin between the current and the historical state, the historical candidate’s log mass, easy confidence, the edited token’s position, prompt length, and design indicators. The comparison adds R_{easy} to these. The outcome is a strict downstream failure on the historical constructions. Models are L2-regularized logistic regressions ($C = 1$, no tuning) evaluated by five-fold cross-validation with folds grouped by history. Out-of-fold log loss with and without R_{easy} is compared by a paired history bootstrap. The downstream answer’s own confidence is reported separately as a contemporaneous reference, not as a predictor, because it comes from the decision query itself.

Protocol history. The confirmation protocol is the second redesign of a design that was frozen before any model was run. The first version scored action events that the model never produces and was corrected before inference. The second failed its competence gate because a trailing Answer : line made the model echo the prefix. The redesign changed only that line, the seeds and the name pools. Appendix A lists every change and when it was made. The protocol records are internal and dated. We call them *frozen before confirmation*, not preregistered.

5 Results

All numbers in this section come from the frozen confirmation (Qwen3-8B, held-out template, 384 histories per task) unless marked otherwise.

5.1 Competence

The routing cells meet all competence criteria; the access cells do not, mainly because current-only

decisions fall short of 0.90 (Table 1). Asked for the *initial* state on the superseded items, the model answered correctly in 0.990 of routing and 0.973 of access cases. In routing, then, the model reports the current state, reports the initial state, and applies the policy without history at or above 0.95. Access results are reported below as fixed-grid averages over cells that do not meet the competence criteria.

5.2 Obsolete states change decisions

H1 is supported in both tasks. When the superseded state implied the wrong action rather than the correct one, the wrong action’s log-odds rose by $D_{\text{superseded}} = 3.904$ nats in access (97.5% interval [3.474, 4.346]) and 3.133 [2.743, 3.516] in routing. For scale, editing the *current* state itself, which legitimately changes the correct action, moved the same quantity by -10.100 and -8.197 nats.

The shift reaches generated answers. The net switch rate toward the wrong action was 0.167 [0.130, 0.206] in access and 0.185 [0.143, 0.227] in routing. Switches were nearly one-directional: 65 histories switched to the wrong action and 1 to the correct one in access, 75 and 4 in routing. In each competent routing cell, the net switch rate was between 0.164 and 0.227 (Table 3). Figure 3 shows this as accuracy. In every routing cell, decisions were correct in 0.906–0.953 of histories when the superseded state implied the correct action, and in 0.727–0.750 when it implied the wrong one. The same records gave current- and initial-state retrieval at or above 0.98.

5.3 The interference is not specific to supersession

H2 compares the superseded assignment with an assignment of the same word to a different attribute that was later updated. It is supported in routing, $\Delta_{\text{semantic}} = +1.224$ [0.973, 1.485], and inconclusive in access, $+0.035$ $[-0.318, 0.400]$. The generated switches agree: $+0.120$ [0.076, 0.164] in routing, $+0.029$ $[-0.016, 0.073]$ in access. Whether the word was assigned to the queried attribute therefore mattered in routing but was not detectable in access.

The mention control, a secondary outcome, gives the clearer answer. When the same entity only *mentioned* the word, the decision effect was larger than for the superseded assignment: $D_{\text{entity mention}} = 5.182$ against 3.904 in access and 4.807 against 3.133 in routing (Figure 2a). That gives $\Delta_{\text{mention}} = -1.278$ $[-1.579, -0.977]$ and

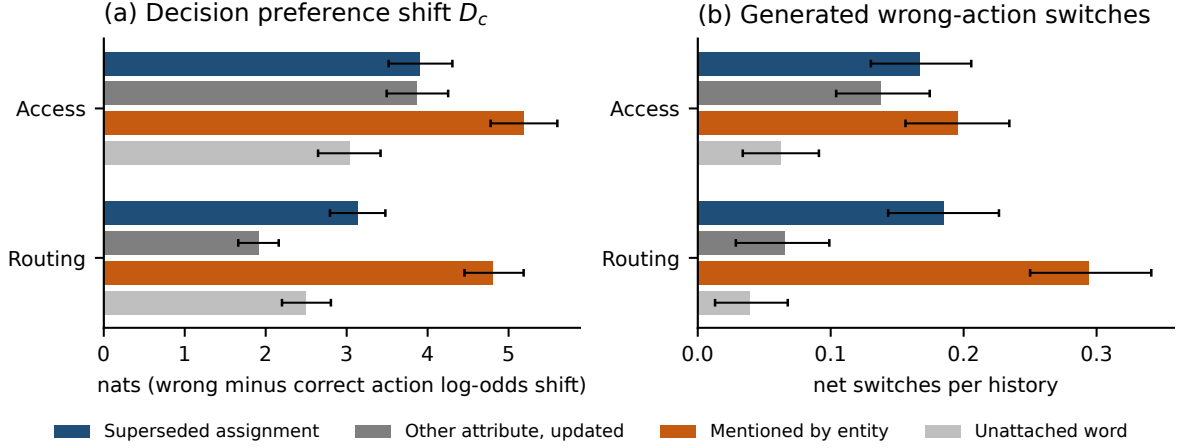


Figure 2: Decision effects by construction (confirmation, 384 histories per task, 95% history-bootstrap intervals). (a) D_c : increase in the wrong action’s log-odds when the edited word implies the wrong rather than the correct action. (b) Net generated switches toward the wrong action per history.

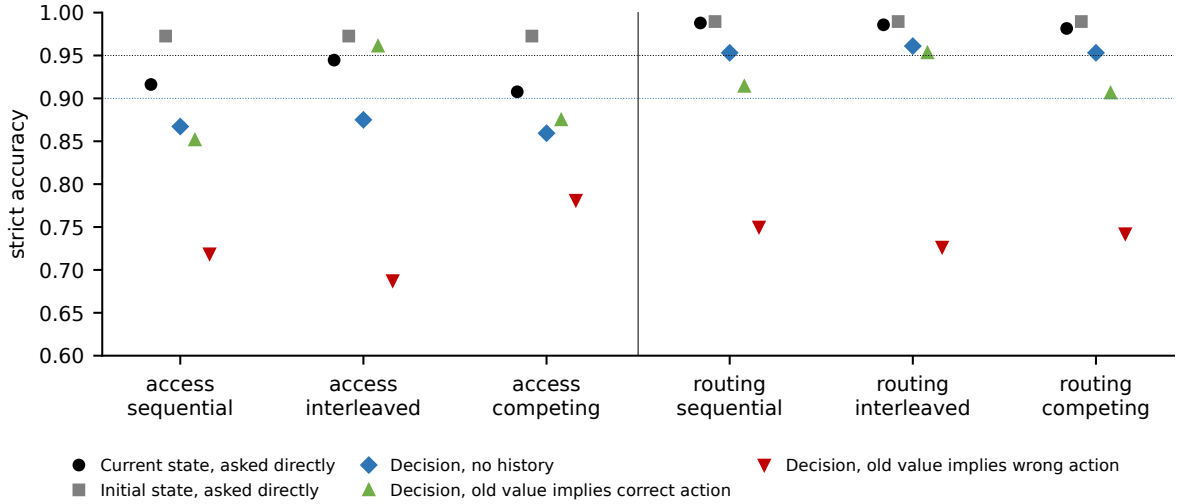


Figure 3: Retrieval and decision accuracy on the same confirmation records, per cell. Initial-state retrieval is reported per task. Dotted lines mark the competence thresholds (0.95 retrieval, 0.90 decisions).

–1.675 [–1.896, –1.441]. In generated answers, mentions caused at least as many switches as superseded assignments: 0.195 (76 to wrong, 1 to correct) against 0.167 in access, an inconclusive difference, and 0.294 (113 against 0) against 0.185 in routing, difference –0.109 [–0.156, –0.065]. A word attached to no entity still shifted preference substantially, by 3.043 and 2.496 nats, but rarely changed a generated decision (0.062 and 0.039). Score shifts of similar size can thus have very different behavioral consequences.

Replication across splits. The same protocol family was also run on four earlier splits: the development and gate splits of the first redesign and of the confirmed protocol. These used the standard

template, and the first redesign kept the Answer: line. They are not confirmatory, but they show which conclusions depend on wording (Figure 4 in Appendix B). $D_{\text{superseded}}$ was positive in all ten split-by-task estimates, between 3.133 and 6.599 nats. Δ_{semantic} was positive with an interval excluding zero in every estimate except the confirmation’s access task. Δ_{mention} never favored the superseded assignment. On the standard template, however, it ranged from –0.806 to +0.163, and six of eight intervals included zero. On that template, mention effects were indistinguishable from or larger than superseded effects. On the held-out template, they were clearly larger. No split showed a superseded assignment exceeding the mention

	Overall	Access	Routing
Failures	892/6144	454/3072	438/3072
Log loss, prevalence	0.414	0.419	0.410
Log loss, pre-decision	0.346	0.323	0.358
AUROC, pre-decision	0.780	0.834	0.729
AUROC, $+R_{\text{easy}}$	0.779	0.833	0.730
Gain from R_{easy}	+0.0004	−0.0006	+0.0013

Table 2: H3: out-of-fold prediction of strict downstream failure on historical-construction items (history-grouped five-fold cross-validation). 95% intervals for the gain: overall $[-0.0011, 0.0019]$, access $[-0.0020, 0.0008]$, routing $[-0.0022, 0.0047]$. The decision’s own action confidence, a contemporaneous reference that is not available before the decision, reaches AUROC 1.000.

control. The size of the mention advantage, however, depends on wording.

5.4 Easy-query sensitivity does not predict failures

The edit strongly affected easy retrieval scores: the mean R_{easy} on superseded items was 8.096 nats in access and 10.878 in routing. That sensitivity carried no information about which decisions would fail (Table 2). Pre-decision features reduced log loss from 0.414 to 0.346 and reached AUROC 0.780. Adding R_{easy} changed log loss by +0.0004 nats, interval $[-0.0011, 0.0019]$; the upper end is under 3% of what the simple features already gain. The result is the same in each task and for the wrong-answer outcome. Within this design, R_{easy} describes how retrieval scores respond to the edit; it does not diagnose decision vulnerability.

5.5 A second model did not meet the competence gate

Gemma 3 4B (Gemma Team, 2025) was added under a model list frozen before any of its data were scored, with the same rows, rules and runtime. The only change was its end-of-turn event, corrected after a format-only smoke test (Appendix A). On the gate split, no cell met the criteria. Current-only decision accuracy was 0.609–0.719, direct retrieval 0.916–0.964, and decision accuracy with history 0.488–0.678. Initial-state retrieval in development was 0.984. Gemma reads these records but does not apply the policy reliably, so no confirmation split was generated for it. This is an eligibility result: it says nothing for or against stale-state interference in Gemma. The conclusions above therefore rest on one model.

6 Discussion

What the evidence supports. In a model that reports both the current and the initial state correctly, an obsolete state still changes about one decision in six toward the action it implies. This shows that stale-value interference, documented for value queries (Guo et al., 2026; Wang and Sun, 2025), also reaches single-step policy decisions in an open model after a single update, in items where retrieval is at ceiling. It is a gap that retrieval accuracy would not reveal. The mention control changes how the finding should be read. The same word, mentioned once by the same entity and never assigned, moves decisions as much or more. In our records, the interference behaves like entrainment by decision-relevant words attached to the queried entity (Niu et al., 2025), not like a failure to track which assignment is current. Supersession contributes only in the routing comparison with an updated unrelated attribute. There it adds 1.2 nats.

Consequences for evaluation and mitigation.

Two practical points follow. First, stale-state benchmarks that vary only update histories cannot tell supersession from incidental mention. A matched mention condition is cheap and changed our conclusion. Second, mitigations that act on update structure, such as removing superseded entries, marking them as historical, or routing attention to the current entry, address only part of the effect if mentions behave the same way. We did not test a mitigation, so this is a prediction, not a result. A selective mitigation would have to be evaluated jointly on decisions and on historical retrieval, which the model performs well and which a blunt fix could destroy.

Scores and behavior. Two score-level results caution against reading log-probability shifts as behavior. An unattached word shifted preferences by 2.5–3.0 nats but flipped few decisions. And retrieval-score sensitivity to the edit, although large, did not predict which decisions failed. A preference shift matters behaviorally only when it crosses the item’s decision margin, so it can be large and still inconsequential. Studies that measure only scores, or only accuracy, see different parts of the same effect.

Alternative explanations. The constructions differ in syntax around the edited word, so the contrasts compare whole constructions. The mention construction could be read as more salient. Its advantage varied with template, which fits a surface

explanation better than a categorical one. The other entity’s current state always implied the wrong action. Confusing the two entities therefore always yields the wrong action. This may contribute to baseline errors, but it cancels in every paired contrast. Natural state words carry prior meanings, but the pattern does not depend on them. In all 12 task $\times$ cell $\times$ vocabulary strata (64 histories each), $D_{\text{superseded}}$ was positive (2.53–4.53 nats). Δ_{mention} was negative in all 12, with intervals excluding zero in 11, including all six with opaque labels (Table 4).

7 Conclusion

In an open model that reports both the current and the initial state of a record correctly, an obsolete state still pulls about one decision in six toward the action it implies. The pull does not depend on the state having been superseded. The same word mentioned in passing by the same entity does at least as much, and how much more depends on wording. A score-level measure of how retrieval reacts to the edit does not identify which decisions will fail. For evaluation, this means stale-context benchmarks need incidental-mention controls and decision-level outcomes, not only update histories and retrieval accuracy. For mitigation, it means fixes aimed at superseded entries are unlikely to be sufficient. Whether these conclusions hold in other models is open. The second model we tested could not perform the base task reliably enough to be evaluated.

Limitations

One model. All confirmed results are for Qwen3-8B. Gemma 3 4B did not meet the competence gate, and no other model was run under this protocol. We make no claim about other models or model sizes.

Synthetic tasks. The records, policies and constructions are synthetic and short: two entities, one update and an explicit policy. This control is what lets the contrasts hold the correct action fixed, but the effect sizes need not transfer to natural dialogue, long agent logs or tool outputs.

Competence in access. The access cells did not meet the competence criteria, mainly because current-only accuracy was 0.859–0.875. Access estimates are fixed-grid averages over cells where the model already makes errors without history.

Construction differences. Constructions share the edited word and its position but differ in syntax. The contrasts compare constructions, not isolated semantic properties. The template dependence of the mention contrast shows that wording matters.

Behavioral evidence only. We measure scores and generated answers. Nothing here identifies a mechanism, and terms such as *entrainment* describe the behavioral pattern, not a verified internal process.

Prediction scope. H3 uses cross-validation within the confirmation split. A null result there does not rule out other score-based diagnostics, other outcomes, or other models.

No mitigation. We did not test an intervention. Statements about mitigation are predictions.

Use of AI assistants. A large language model assisted with implementing the experiment code and drafting this manuscript. The authors checked the claims, analyses, references and final text.

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A Protocol history

Every change below is recorded, with its date, in the protocol documents released with the code. None was made after looking at confirmation data.

v1 (frozen design, never run). The design fixed the hypotheses, estimands, factorial, constructions, gates, bootstrap and held-out confirmation template. An audit before any inference found two problems. (i) The action events began with a space (`_APPROVE`). Qwen3-8B begins its turn without one, and the two strings tokenize differently, so v1 would have scored continuations the model never produces. (ii) Among the H3 predictors was the decision’s own action confidence, which is not available before the decision is made.

v1b (corrected before inference). Action events became the label exactly as generated plus the end-of-turn token. The H3 predictors were split into pre-decision predictors and a contemporaneous reference. Historical retrieval was added as a competence measure. During setup, the runtime was changed twice for numerical reasons (Appendix A.1), and a rounding tolerance of 10^{-5} was added to validators that rejected float32 log probabilities a few 10^{-7} above zero. The v1b gate failed: direct retrieval was 0.851–0.946 in every cell. The cause was the prompt format. The user message ended with a literal `Answer: line`, which the model often echoed (about 260 of 4,224 retrieval answers); about 100 answers gave an action label instead of a state. Only 64 named the obsolete value. The decision effects replicated on the gate histories.

v2 (first redesign; at most two were allowed). The redesign changed only what the failure required: it removed the trailing `Answer: line` and used new seeds and entity-name pools so that all histories were new. Confirmation size was set to four replicates (768 histories) before any v2 data existed, to fit the compute budget. The scorer was

changed from a batched scorer to a key-value-cache scorer for speed, before any v2 effect was inspected. Both scorers compute the same quantity, and the cache scorer passed its equivalence check. The v2 gate passed in three cells, and confirmation data were then generated.

Second model. The model list, adding Gemma 3 4B with the same rows, rules and runtime, was fixed before any Gemma data were scored. A 24-record smoke test, inspected for format only, showed that Gemma writes its label, a newline and then its end-of-turn token. The scored events had omitted the newline and covered at most a few percent of the probability mass. The scored event was corrected to include the newline, the checks were rerun, and the smoke outputs were set aside. Development and gate then failed the competence criteria in every cell (§5.5).

A.1 Numerical causality of log-probability scoring

Scoring answers by teacher forcing assumes that the log probability at a position does not depend on tokens appended after it. We tested this directly on the same prompts, by comparing the next-token distribution at a fixed position with and without one or two tokens appended. With 8-bit weights (LLM.int8(), Dettmers et al., 2022), log probabilities changed by up to 0.843 nats. With bfloat16, the top-20 log probabilities changed by up to 0.875 nats, and computing only the output layer in float32 reduced it to 0.502. In float32, and with 4-bit nf4 weights, it was exactly zero. In practice this broke our validators: four disjoint candidate answers summed to more than probability one. A key-value-cache scorer also disagreed with full re-computation by 3.406 nats under 8-bit weights, but by 5.08×10^{-5} in float32. Score differences of a few nats are the size of the effects studied here, so we ran all inference in float32. Score-based studies that use reduced precision should report a check of this kind.

B Additional results

C Reproducibility

Models. Qwen3-8B, Hugging Face revision b968826d9c46dd6066d109eabc6255188de91218; Gemma 3 4B (instruction-tuned), revision 093f9f388b31de276ce2de164bdc2081324b9767. Both ran in float32 without quantization, with

Cell	Superseded	Other attr.	Mention	Unattached
access seq.	0.133	0.070	0.148	0.094
access int.	0.273	0.234	0.219	0.094
access comp.	0.094	0.109	0.219	0.000
routing seq.	0.164	0.062	0.352	0.055
routing int.	0.227	0.078	0.273	0.055
routing comp.	0.164	0.055	0.258	0.008

Table 3: Net generated switch rate toward the wrong action, per cell and construction (confirmation, 128 histories per cell). Intervals are in the released analysis.

Stratum	Vocab.	$D_{\text{superseded}}$	Δ_{mention}
access seq.	natural	3.61	-1.16
access int.	natural	4.01	-0.06
access comp.	natural	3.65	-2.21
access seq.	opaque	4.53	-1.10
access int.	opaque	4.43	-1.42
access comp.	opaque	3.20	-1.72
routing seq.	natural	3.66	-2.32
routing int.	natural	4.03	-1.00
routing comp.	natural	3.07	-2.38
routing seq.	opaque	2.53	-2.17
routing int.	opaque	2.78	-1.32
routing comp.	opaque	2.73	-0.87

Table 4: Decision effects (nats) by task, cell and vocabulary (confirmation, 64 histories per stratum). All $D_{\text{superseded}}$ intervals exclude zero. The Δ_{mention} interval includes zero only for access, interleaved, natural ($[-1.01, 0.96]$).

SDPA attention, on one 80 GB H100 GPU, using Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019).

Artifacts. Each dataset is regenerated deterministically from its seed and checked against its manifest. Each score file carries per-record checksums and a completion record binding the scores, retrieval scores, run manifest, tokenizer audit, code hash and configuration hash. Analyses refuse incomplete or altered runs. Confirmation: seed 82003, dataset hash 3e972257. . . , code hash c6648aba. . . , configuration hash 78d09215. . . , score file SHA-256 b159b727. . . . The protocol was frozen on 2026-10-08 (UTC), before the gate and confirmation splits were generated. Every number and figure in this paper is produced from the sealed analysis reports by one script, which records the hash of each input.

Compute. Confirmation scoring took 6.27 GPU hours (2.67 s per record, including two retrieval queries, one historical query where applicable, two scored action events and greedy generation). The

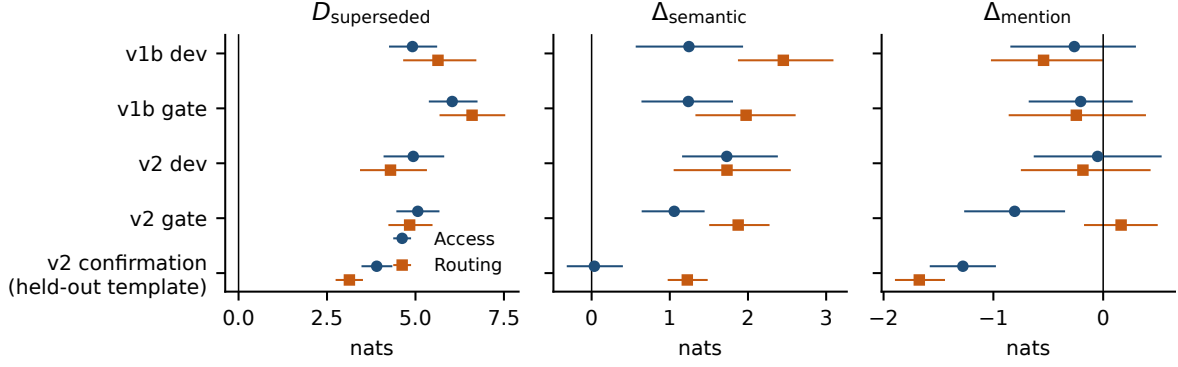


Figure 4: Primary and mention contrasts across all five splits run under the v1b and v2 protocols (95% intervals for development and gate splits, 97.5% for the confirmation’s primary contrasts and 95% for its Δ_{mention}). Development and gate splits used the standard template, and v1b also kept the Answer : line. Only the confirmation is confirmatory.

Cell	Direct	No history	History
access seq.	0.945	0.719	0.678
access int.	0.936	0.609	0.676
access comp.	0.946	0.672	0.641
routing seq.	0.939	0.688	0.645
routing int.	0.964	0.609	0.488
routing comp.	0.916	0.625	0.533

Table 5: Gemma 3 4B on the frozen gate split (4,224 records, the same rows as the Qwen3-8B gate). No cell meets the competence criteria. No answer was invalid.

whole program, including failed runs, development and gate splits and the second model, used about 24 GPU hours.

Code. [TODO: anonymous repository link] contains the generators, scoring and analysis code, tests, configuration files, protocol records with their dated amendments, and the derivation and figure scripts.